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Statement of Research

Postdoctoral Researcher, Foundations of a Resilient Microbiome (JPF03176)

Overview

My research asks how microbial communities, and the elements that live within them such as phages, plasmids, and the peptides they encode, shape the health of their hosts. I have pursued this question across very different systems, from children with obesity to farmed shrimp and laboratory mice, and that variety has become one of my greatest strengths. The common thread is a steady movement from describing which microbes are present toward understanding what communities actually do. What I want to understand next is why some communities recover after a disturbance while others do not, and whether we can predict it.

Research to date

From composition to function. Much of my PhD at UNAM's Institute of Biotechnology focused on what the gut microbiota of Mexican children with obesity and metabolic syndrome is actively producing. In my first-author work, I defined the "Secrebiome," the set of proteins that gut bacteria secrete into their environment, and my dissertation, *Identification of Antimicrobial Peptides Expressed by the Gut Microbiota Associated with Obesity and Metabolic Syndrome*, identified antimicrobial peptides encoded by plasmids and phages and explored their association with metabolic disease. This work showed me that microbial communities carry their own tools for competition and defense, and that these tools are likely part of what keeps a community stable, or pushes it out of balance.

The virome as a hidden driver. In parallel, I contributed to studies showing that the gut virome changes in childhood obesity and that different groups of crAssphage, among the most abundant phages in the human gut, have opposing associations with metabolic syndrome. I also co-developed a published protocol for studying the human gut phageome. These projects convinced me that phages are not passive passengers but active forces that shape bacterial populations, and that any realistic picture of microbiome resilience needs to include them.

Hosts, environments, and interventions. Beyond the human gut, I have studied how host genetics and the environment together shape the microbiota of farmed shrimp, and how environmental enrichment slowed the progression of gut dysbiosis and improved glucose metabolism in obese mice. That mouse study is, in essence, a resilience experiment: a change in the environment altered the trajectory of a disrupted community. I have also worked on the genomes and transcriptomes of non-model organisms, from probiotic bacteria to tropical fruit. Moving between these systems, and between the bench and the computer, taught me to adapt quickly to new questions and collaborators, and to recognize the patterns that hold across very different organisms.

What I hope to contribute

Bringing microbial functions into models. Models of community dynamics are often built on who is present. I would like to help enrich them with what communities can do, including phage activity and antimicrobial or bacteriocin potential, which plausibly

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mediate competition and stability. Integrating these functional features with metagenomic and metabolomic measurements from ExFAB could help explain why some communities withstand disruptions such as antibiotics while others do not. Most of my experience lies in turning large multi-omics datasets into interpretable results, and I would apply it to building statistical and machine learning models whose predictions can be tested directly in ExFAB.

Connecting early-life disruptions to long-term trajectories. My work in childhood obesity made me appreciate how much early-life microbial communities may matter for lifelong health. I am eager to bring that perspective to the gut-brain axis, helping to identify which community states and metabolic signals are linked to divergent developmental trajectories, and to work with the team to separate associations from plausible causal mechanisms.

Closing the loop between models and experiments. What excites me most about this project is that models and experiments inform each other. Because I have worked at both ends of the research process, I enjoy translating experimental questions into analyses that can be tested, and model predictions into experiments that are practical to run. I am eager to deepen my training in ecological theory, dynamical modeling, and causal inference alongside Professors Carlson and Moeller and their colleagues across physics, ecology, engineering, and psychology.

Long-term goals

In the long term, I want to help develop functional probiotics that actually work across the different stages of life, from infancy to adulthood. Many probiotics are chosen without considering the community they enter or the life stage of the host, and their effects are often modest or inconsistent. My work has approached this question from several angles: contributing to the genomic characterization of a bacterial strain with probiotic potential and to a study showing that the dietary fiber agavin enriched beneficial microbes in the shrimp gut, and identifying antimicrobial peptides that could help beneficial microbes compete. Predictive models of community resilience, like those this project will build, are what is needed to know which strains and functions can truly establish and help at each stage of life, and I see this position as the foundation for that research path.

I also value teaching and open science: I organized and taught the “Bioinfo del Norte” bioinformatics and metagenomics training at CIAD, have given guest lectures in Mexico and Colombia, and directed an undergraduate thesis on phages in childhood obesity, and I look forward to continuing to mentor students and share reproducible tools as part of the UCSB community.